

Sales Force Automation (SFA)

Master Technical & Operational Documentation | Complete System Architecture & Reference Guide

Sales Force Automation (SFA) — Project Master Documentation

NOTE: This master document provides a technical and operational overview of the **Sales Force Automation (SFA)** platform. It covers system architecture, role-based access control (RBAC), end-to-end business workflows, database models, target/performance frameworks, and analytics modules.

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1. Executive Summary & System Purpose

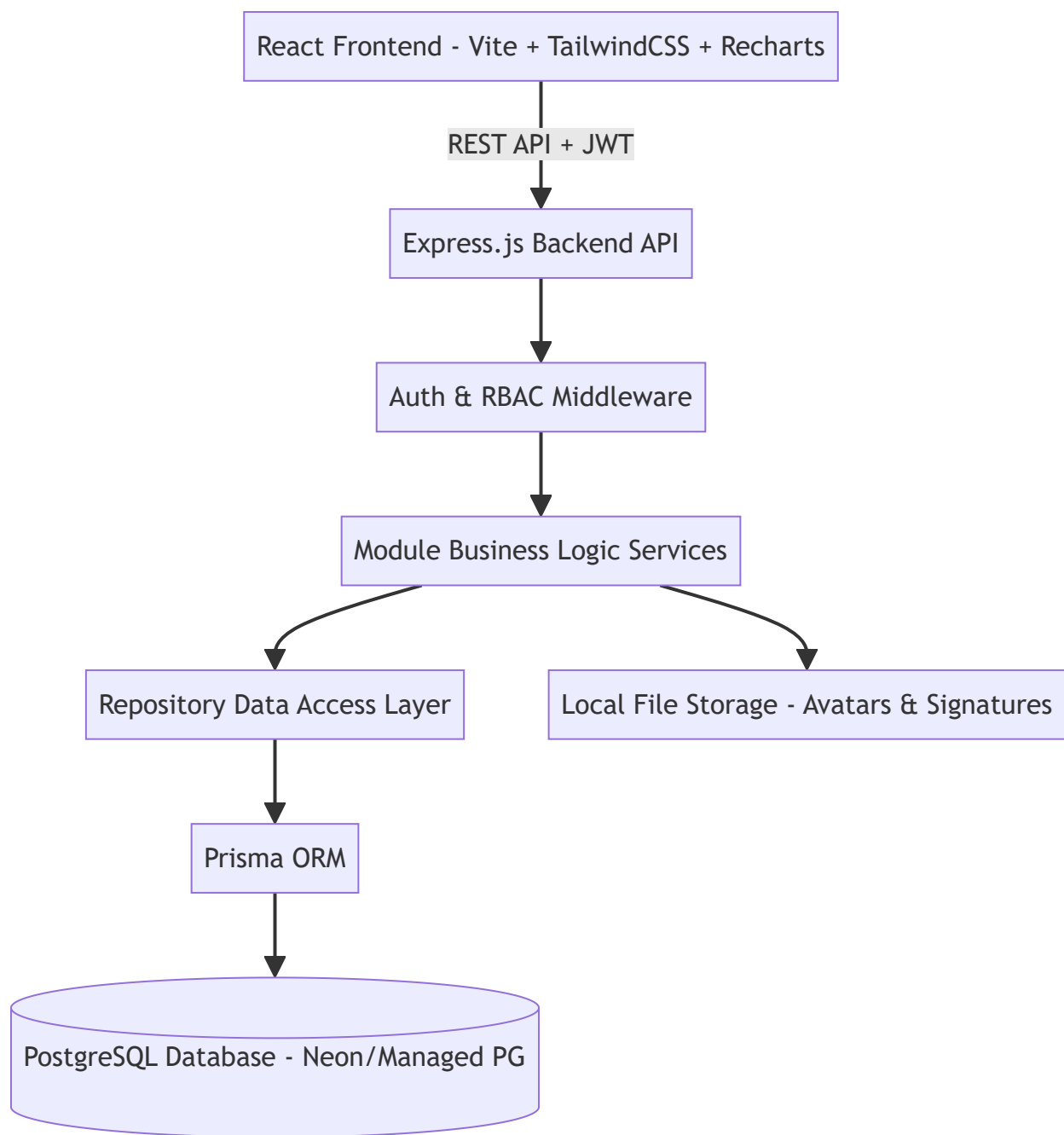
The **Sales Force Automation (SFA)** platform is an operational-sales management system designed for multi-branch distribution, field force tracking, inventory logistics, and order fulfillment.

IMPORTANT: Business Focus Directive: SFA is an **operational-sales system**, not a financial accounting software. Primary performance indicators focus on operational throughput (Tasks Completed, Orders Fulfilled, Workforce Efficiency) alongside real-time sales order revenue tracking.

Key Capabilities

- **Multi-Tenant & Multi-Branch Architecture:** Isolated data scopes per Organization, Branch, Warehouse, and Team.
- **Field Force Automation:** Real-time task tracking, customer visit logging, digital signature capture, and geofence-assisted handovers.
- **Warehouse & Inventory Integration:** Multi-warehouse stock tracking, stock movements, reserved vs. available quantities, and automated handover approvals.
- **Operational Target & Performance System:** Dynamic operational performance tracking for Sales Executives (Task Execution %) and Sales Managers (Order Fulfillment %).
- **Visualization-First Reports:** 0-export interactive Recharts analytics describing Branch Performance, Field Force Workforce, Warehouse Stock, and Sales Revenue Trajectory.

2. Technology Stack & System Architecture



Stack Breakdown

- **Frontend:** React (Vite), TailwindCSS, Recharts for dynamic visual graphs, Lucide React icons, Framer Motion for micro-animations.
- **Backend:** Node.js, Express.js (Modular Monolith Architecture), Prisma ORM (Client v7+).
- **Database:** PostgreSQL (Hosted on Neon / Managed Postgres) with SSL pooling.
- **Authentication:** JWT-based Bearer Authentication with password hashing via bcryptjs.
- **File Storage:** Local uploads directory with static express routing for user avatars and digital signatures.

3. Role-Based Access Control (RBAC) Matrix

The system enforces strict role-based scoping across 7 distinct roles:

Role Name	Access Level	Primary System Focus
Organization Super Admin	Platform-Wide	Global organization oversight, Company Admin management, multi-branch analytics.
Company Admin	Organization Level	Branch, Department, Team, and User account administration.
Head of Sales	Strategic Sales Level	Revenue trajectory, Target & Performance dashboard, company-wide operational reports.
Sales Manager	Branch Level	Branch operational management, team task assignment, branch sales order completion.
Sales Executive	Field / Individual Level	Task execution, customer visits, digital signature capture, sales order placement.
Inventory Manager	Global Logistics	Product catalog, multi-warehouse stock management, global stock transfers.
Warehouse Manager	Warehouse Console	Warehouse stock check-in/check-out, pickup handover approvals, stock movements.

4. Core Business Workflows & Lifecycles

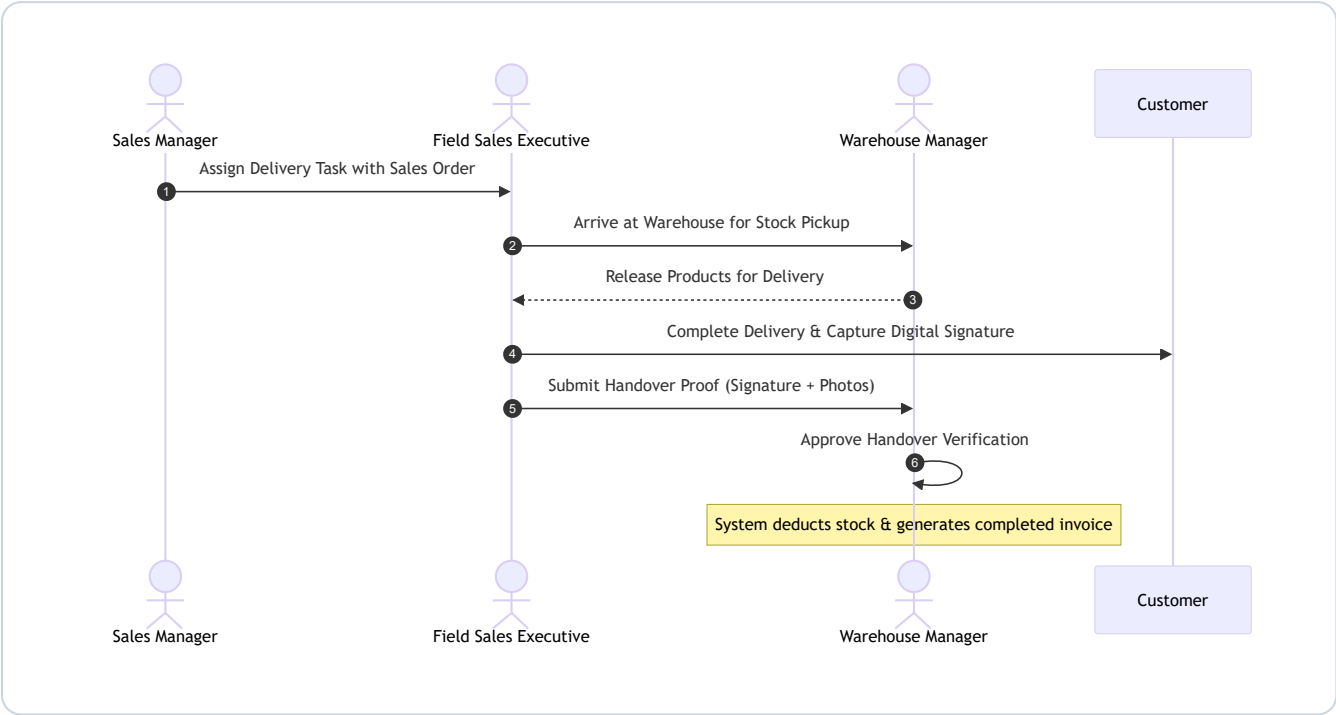
A. Sales Order Lifecycle



- **Order Creation (DRAFT)**: Created by Sales Executive or Sales Manager with selected customer, line items, quantities, and prices.
- **Order Submission (SUBMITTED)**: Order lock for approval review.
- **Approval & Allocation (APPROVED)**: Verified against stock availability in designated warehouses.

- **Fulfillment & Delivery** (`IN_FULFILLMENT` $\rightarrow$ `DELIVERED`): Field executive picks up items and delivers to customer.
- **Invoice & Completion** (`COMPLETED`): Customer signs digital delivery proof; Warehouse Manager approves handover, automatically triggering stock deduction and invoice generation.

B. Task Execution & Warehouse Handover Flow

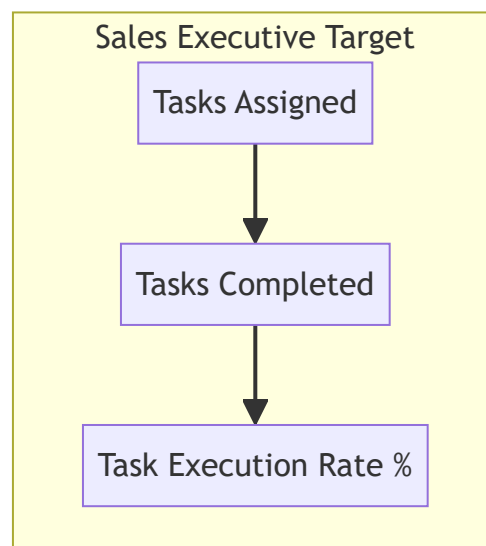
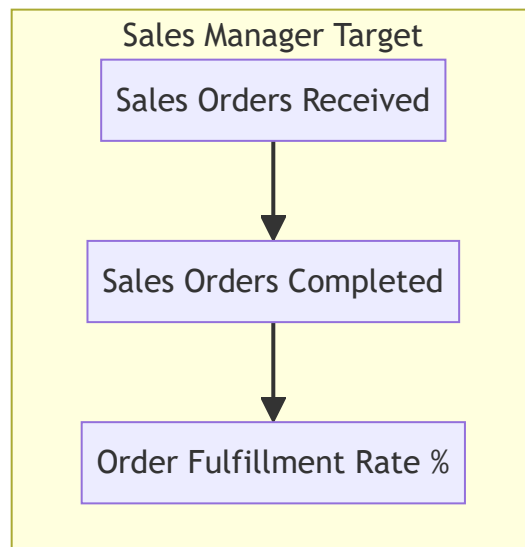
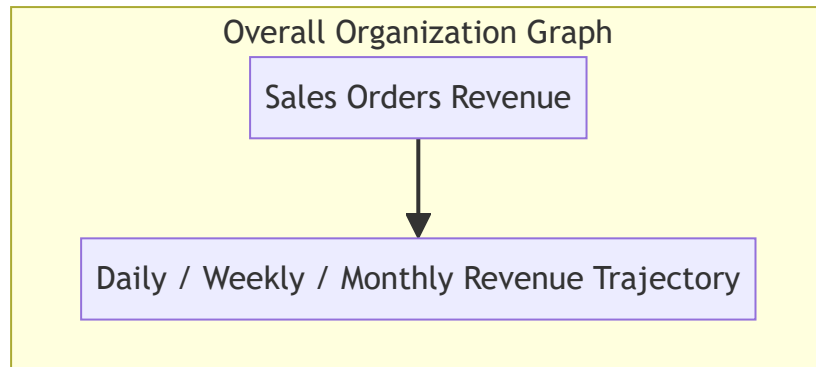


C. Inventory & Stock Movement Flow

- **Net Available Quantity Calculation:**
$$\text{Net Available Qty} = \text{Total Physical Quantity} - \text{Reserved Quantity}$$
- **Stock Movement Types:**
 - `CHECK_IN` : Initial stock load or purchase receipt into warehouse.
 - `CHECK_OUT` : Manual stock adjustment or write-off.
 - `TRANSFER_IN` / `TRANSFER_OUT` : Inter-warehouse or branch stock transfers.
 - `FULFILLMENT_DEDUCTION` : Automated stock deduction upon completed sales order handover.

5. Target & Operational Performance Module

The **Target & Performance** module evaluates users strictly by **Operational Metrics** appropriate to their role:



Role-Based Performance Metrics

- **Sales Executives:**
- **Target Indicator:** Tasks Assigned vs Tasks Completed.

- **Formula:** $\text{Task Execution \%} = \frac{\text{Completed Tasks}}{\text{Total Assigned Tasks}} \times 100\%$
 - **Sales Managers:**
 - **Target Indicator: Sales Orders Placed/Received vs Sales Orders Completed.**
 - **Formula:** $\text{Order Completion \%} = \frac{\text{Completed Orders}}{\text{Total Orders Placed}} \times 100\%$
 - **Organization Revenue Trajectory:**
 - Dynamic Area & Line charts plotting real Sales Order Revenue (₹) over **Daily**, **Weekly**, and **Monthly** periods.
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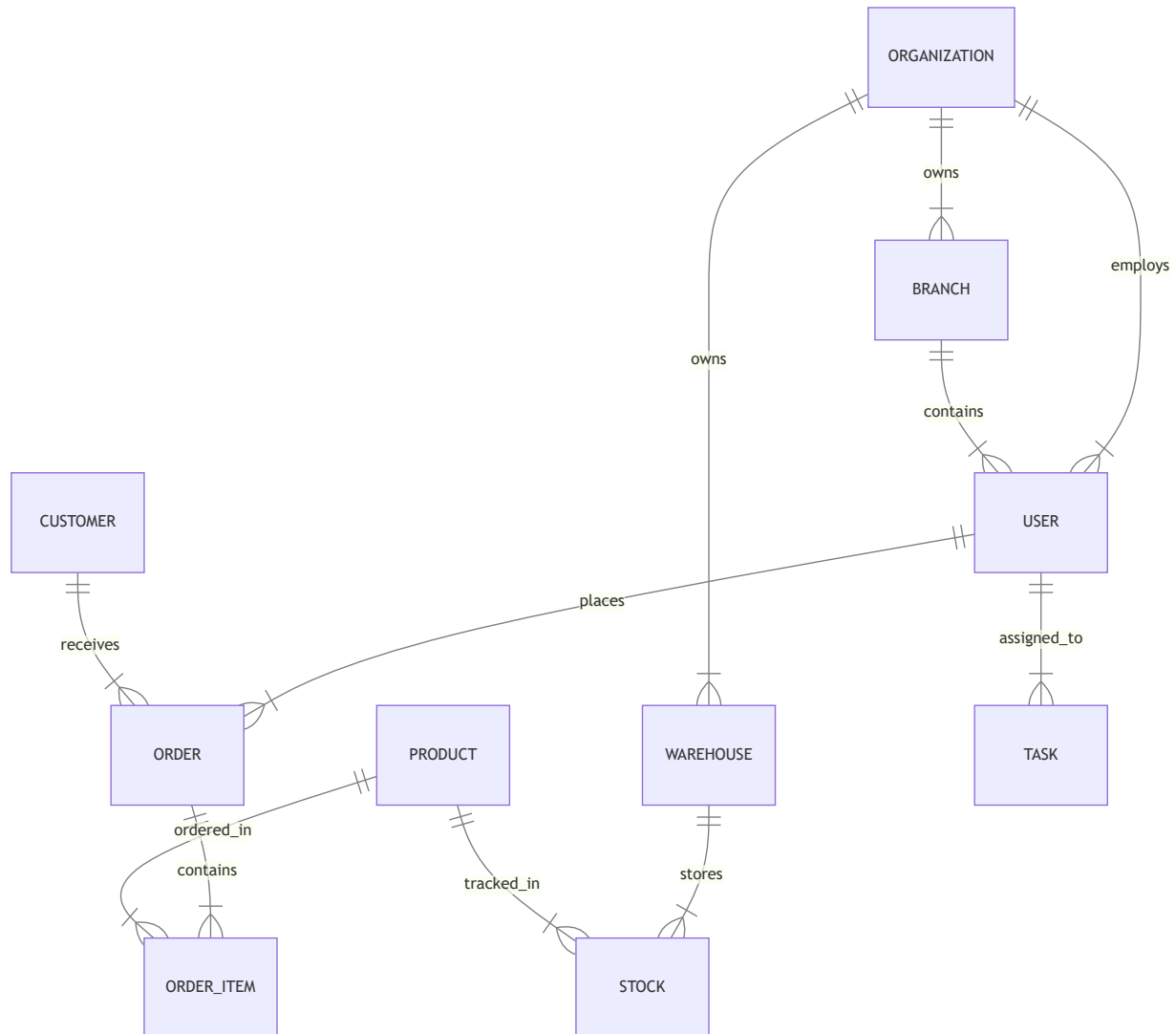
6. Reports & Business Intelligence Module

The **Reports Page** is designed with **0 export/download buttons** to provide an immediate, visual-first executive dashboard across 5 core tabs:

- **Branch Operational Performance:**
 - Real DB counts for `Total Orders` , `Completed Tasks` , `Requested Quantity` , `Staff Count` per branch.
 - **Warehouses Report:**
 - Total warehouses, capacity utilization %, total stock value, and warehouse manager assignments.
 - **Product Report:**
 - Product catalog metrics, price list, category breakdown, and total quantities sold.
 - **Stock per Branches Report:**
 - Table detailing Product, SKU, Warehouse, Branch Assignment, and **Net Available Quantity**.
 - **Field Force Workforce & Task Execution Report:**
 - Field workforce count, overall task completion rate %, and per-worker task breakdown table (Worker Name, Email, Branch, Assigned Tasks, Completed Tasks, Pending Tasks).
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7. Database Schema & Data Models

Core Entities & Relationships



Primary Models (`prisma/schema.prisma`)

- **Organization** : Multi-tenant root entity.
- **Branch** : Physical business locations linked to users, orders, and warehouses.
- **User** : Account entity linked to `UserRole`, `Branch`, `Department`, `Team`. Includes `avatarUrl` for profile storage.
- **Warehouse & Stock** : Logistics inventory entities tracking physical quantity, reserved quantity, and location coordinates.
- **Order & OrderItem** : Sales order entities tracking customer, owner, status, line items, and total amount.
- **Task** : Field delivery task linked to assigned worker, pickup location, delivery location, customer signature, and status (`PENDING`, `IN_PROGRESS`, `COMPLETED`).
- **Target** : Operational performance target records for users and teams.

8. API Architecture & Route Sitemap

API Endpoints Overview

Base Endpoint	Method	Description
/api/v1/auth/login	POST	User login & JWT issuance
/api/v1/auth/me	GET	Current authenticated user profile
/api/v1/users	GET , POST	User management (Supports <code>limit=1000</code> for full org list)
/api/v1/users/profile	PUT	Update user profile & avatar upload
/api/v1/sales-orders	GET , POST	Sales order CRUD & lifecycle updates
/api/v1/target-performance/company-overview	GET	Target & performance operational analytics & trend graphs
/api/v1/reports/organization-analytics	GET	5-tab operational reports payload
/api/v1/inventory/warehouses	GET , POST	Warehouse management
/api/v1/inventory/stock	GET , POST	Stock level tracking & movements
/api/v1/team/tasks	GET , POST	Task creation, status updates & signature handover

9. User Settings & Profile Persistence

Avatar / Profile Picture Flow

- **Upload:** User selects image in Settings Page (`SettingsPage.jsx`).
- **Storage:** Backend saves file to `/uploads/avatars/avatar-[userId]-[timestamp].[ext]` .
- **Database:** Updates `User.avatarUrl` column in PostgreSQL database.
- **Auth Sync:** `AuthContext.jsx` updates local storage state so the avatar persists seamlessly across page refreshes and sessions.

TIP: Verification Status: All modules described in this documentation have been verified against the live PostgreSQL database and production Vite build (`npm run build` completed in **1.94s** with 0 errors).